



Precedent Results

Objective

¿En qué medida la literatura científica actual ha explorado arquitecturas híbridas de aprendizaje profundo —que integren codificadores espectrales basados en atención o grafos (Transformer/GNN), modelado temporal eficiente (LSTM, GRU o modelos de espacio de estados tipo Mamba) y restricciones cinéticas informadas por física (PINN/Neural ODE)— para predecir la degradación temporal de propiedades fisicoquímicas a partir de espectros FTIR en alimentos de alto valor comercial como cacao, café, miel y aceite de coco, y qué vacíos metodológicos persisten respecto a la integración multi-matriz y la interpretabilidad química de dichos modelos?

Results

No. # Estado de la literatura

Conclusión ejecutiva

La literatura científica ha explorado **los componentes de la arquitectura propuesta, pero casi siempre por separado**. En las búsquedas realizadas no se identificó ningún trabajo que combine, sobre series longitudinales FTIR de cacao, café, miel o aceite de coco, los tres niveles siguientes: **(i)** codificación espectral mediante Transformer o GNN, **(ii)** evolución temporal mediante LSTM, GRU o Mamba y **(iii)** regularización mediante una PINN o Neural ODE que imponga cinética fisicoquímica. Por tanto, el grado de exploración de la arquitectura completa debe calificarse como **incipiente y fragmentario**, no como una línea ya establecida.

Lo que sí existe es: FTIR alimentario con atención/Transformer para clasificación estática; seguimiento espectroscópico de procesos con quimiometría; modelos recurrentes aislados; Mamba espectral para series temporales no alimentarias; y modelos informados por física para secado o tostado sin entrada FTIR. La convergencia de esas líneas constituye todavía un vacío de investigación.

Evidencia / ámbito	FTIR y matriz	Codificador espectral	Módulo temporal	Restricción física / cinética	Tarea real	Conclusión / vacío
Cacao	FT-MIR/FTIR se usa principalmente para calidad, autenticidad, origen y adulteración	Predominan PCA, PLS/PLS-DA, SVM, RF y MLP; Transformer/GNN no identificado	LSTM/GRU/Mamba no identificado	PINN/Neural ODE no identificado	Clasificación o regresión estática de composición, calidad, origen o adulteración	No se identificó predicción longitudinal de degradación ni una arquitectura híbrida completa. La fusión en cacao emplea principalmente LC-MS/GC-MS y clasificadores convencionales, no FTIR cinético ([1.1], [2.1])

Evidencia / ámbito	FTIR y matriz	Codificador espectral	Módulo temporal	Restricción física / cinética	Tarea real	Conclusión / vacío
Café	Sí: FTIR para autenticación/adulteración; NIR/FT-NIR para composición, calidad y seguimiento del tostado	Principalmente PCA, PLS, RF y CNN; Transformer/GNN aplicado a degradación FTIR no identificado	LSTM/GRU/ Mamba sobre secuencias FTIR longitudinales no identificado	Existe PIML alrededor de una ODE mecánica de transferencia de calor y masa en tostado, pero sin FTIR ni codificador espectral	Autenticación, adulteración, predicción composicional y control del tostado. El estudio PIML modeló 221 tostados y alcanzó R^2 de 0.70–0.94; una MLP empírica obtuvo 0.97	Es la matriz más cercana a una integración físico-temporal, pero los bloques permanecen separados. Falta enlazar espectros FTIR, estados químicos y cinética; tampoco se validó la transferencia entre tostadores, recetas y variedades ([2.2], [3.1], [4.1])
Miel	FTIR/NIR se emplean sobre todo para autenticación, origen botánico, detección de jarabes y composición	Quimiometría y ML convencional; Transformer/GNN no identificado	LSTM/GRU/ Mamba no identificado	PINN/Neural ODE no identificado	Clasificación de origen o adulteración y regresión composicional, generalmente con observaciones independientes	No se identificaron trayectorias espectrales de envejecimiento modeladas dinámicamente ni integración con ecuaciones de cristalización, fermentación, pérdida enzimática o formación de HMF ([2.2])
Aceite de coco / aceites comestibles	FTIR está establecido para clasificación, adulteración, ácidos grasos y monitorización de oxidación; la evidencia específica sobre coco se concentra en autenticidad/adulteración	PCA/PLS y ML convencional; Transformer/GNN FTIR cinético no identificado	LSTM/GRU/ Mamba no identificado	PINN/Neural ODE no identificado	Cribado de autenticidad, cuantificación de adulterantes y evaluación estática o por tiempos separados de oxidación	No se identificó una arquitectura que aprenda simultáneamente espectro, trayectoria y cinética de oxidación. Persisten dependencia de bases de referencia, efectos instrumentales y falta de validación interlaboratorio, estacional y entre aceites ([5.1], [5.2], [5.3])
Antecedente adyacente: FTIR de ghee con atención/Transformer	Sí, FTIR de ghee; matriz distinta de las cuatro solicitadas	CNN con atención y Transformer, fusionados con RF, SVM y XGBoost; mapas de atención espectral	No identificado	No identificado	Clasificación de adulteración; atención sobre bandas como ésteres cerca de 1740 cm^{-1} y estiramientos C–H cerca de 2850 cm^{-1}	Demuestra que un Transformer puede codificar FTIR alimentario y resaltar regiones químicas, pero no modela degradación, tiempo ni cinética. La validación externa ante muestras de campo y adulterantes desconocidos sigue limitada ([6.1], [6.2], [6.3])
Antecedente adyacente: Mamba espectral genérico	No FTIR; ninguna de las cuatro matrices	Compuerta espectral adaptativa basada en energía local de parches antes del SSM	Mamba/SSM con complejidad lineal $O(L)$, frente a atención $O(L^2)$	No identificado	Pronóstico multivariado genérico de series temporales	Aporta eficiencia y filtrado de ruido para horizontes largos, pero no evidencia validez para espectros vibracionales, degradación alimentaria o parámetros cinéticos; sus <i>node embeddings</i> no constituyen una GNN química ([7.1])
Antecedente adyacente: PINN de secado	No FTIR; las cuatro matrices no están identificadas	Red neuronal convencional; sin Transformer/GNN espectral	La dinámica se representa mediante campos de temperatura y humedad, no mediante LSTM/GRU/ Mamba	Modelo mecanístico de balances de calor y masa usado para generar datos y comparar NN convencional con PINN	Simulación y predicción del secado	Prueba que la física puede regularizar procesos dinámicos, pero no conecta bandas FTIR con especies químicas ni valida cinéticas específicas de cacao, café, miel o coco ([8.1])

Evidencia / ámbito	FTIR y matriz	Codificador espectral	Módulo temporal	Restricción física / cinética	Tarea real	Conclusión / vacío
Balance de integración completa	No identificado un estudio longitudinal FTIR en alguna de las cuatro matrices que reúna todos los bloques	No identificado Transformer/GNN químicamente estructurado dentro de la arquitectura completa	No identificado LSTM/GRU/Mamba acoplado al codificador FTIR y a la física	No identificado PINN/Neural ODE con leyes de Arrhenius, oxidación, hidrólisis, fermentación u otras cinéticas específicas	La literatura estudia componentes aislados y tareas mayormente estáticas	Grado de exploración incipiente y fragmentario : no debe confundirse fusión de clasificadores con integración multi-matriz, atención con causalidad química ni muestreo en varios tiempos con modelado dinámico mecanístico ([6.4], [1.2], [9.1], [6.5])

Table: La tabla contrasta la evidencia por matriz y por componente arquitectónico. Muestra que los codificadores espectrales, modelos temporales y restricciones físicas existen como líneas separadas, pero no como una arquitectura FTIR híbrida completa para degradación longitudinal.

Evidencia por bloque arquitectónico

1. Codificación espectral con atención o grafos

El antecedente más próximo es un sistema de autenticación de ghee que combina CNN con atención, Transformer y clasificadores convencionales. Sus mapas de atención resaltan bandas asociadas a enlaces éster alrededor de 1740 cm^{-1} y estiramientos C–H cerca de 2850 cm^{-1} . Esto demuestra que la atención puede localizar regiones FTIR plausibles químicamente, pero la tarea es clasificación de adulteración, no pronóstico de degradación, y no incluye un modelo temporal ni restricciones cinéticas ([6.1], [6.2]).

Los trabajos recuperados sobre cacao, café y miel utilizan principalmente PCA, PLS/PLS-DA, métodos discriminantes, random forest, SVM, MLP y, ocasionalmente, CNN. En café se encuentran tareas de autenticidad, cuantificación de mezclas, composición y seguimiento del tostado; en cacao, de origen, adulteración, composición y calidad; y en miel, de origen o adulteración. Estos antecedentes no implementan la arquitectura híbrida solicitada ([2.2], [1.1], [2.1]).

La **GNN es el bloque menos explorado** en este contexto. No se identificaron grafos cuyos nodos representen bandas, grupos funcionales, especies químicas o indicadores fisicoquímicos y cuyas aristas codifiquen acoplamientos de reacción o correlaciones espectrales. Debe distinguirse además un *node embedding* de variables de una GNN química: ASGMamba utiliza identificadores por variable, pero no realiza propagación sobre un grafo molecular o cinético ([7.1]).

2. Modelado temporal

La mayor parte de los estudios trata cada espectro como una observación independiente. Incluso cuando se toman medidas durante almacenamiento, fermentación o procesado, suelen analizarse mediante PCA, ASCA, PLS o regresiones por punto temporal. Por ejemplo, se ha seguido una fermentación de aceitunas durante 145 días con MIR/Raman y métodos multivariantes, y se han combinado imágenes RGB/NIR, pH y pérdida de peso durante almacenamiento; ello aporta información temporal, pero no equivale a aprender una transición dinámica latente ([1.2]).

Se localizaron antecedentes CNN–LSTM en espectroscopia infrarroja alimentaria, pero para identificar pasta precocida en distintos estados físicos, no para estimar una trayectoria de degradación ni parámetros cinéticos. La evidencia recuperada tampoco mostró GRU aplicada a series FTIR longitudinales de las cuatro matrices ([9.1]).

Mamba ofrece una justificación computacional atractiva: ASGMamba contrasta la complejidad lineal $\mathcal{O}(L)$ de su SSM con la complejidad cuadrática $\mathcal{O}(L^2)$ de la atención, e introduce compuertas espectrales locales para filtrar ruido sin perder ese escalado lineal. Sin embargo, se trata de pronóstico multivariado genérico; no usa FTIR ni alimentos y no demuestra que Mamba preserve picos estrechos, desplazamientos de banda o relaciones estequiométricas en espectros vibracionales ([7.1]).

3. Restricciones físicas y cinéticas

Existen pruebas de concepto relevantes, pero desacopladas de FTIR. Perré combina balances mecánicos de calor y masa con redes neuronales para secado y compara una red convencional con una PINN. El trabajo muestra la utilidad de la física para extrapolar fuera del dominio puramente empírico, aunque no estudia las cuatro matrices ni relaciona bandas FTIR con especies o propiedades químicas ([8.1]).

El antecedente más cercano por matriz es un estudio de 221 tostados industriales de café. Cuatro estrategias de aprendizaje informado por física reparan un modelo ODE mecánico y elevan el desempeño de despliegue autónomo a R^2 de 0.70–0.94. No obstante, una MLP puramente empírica alcanzó 0.97 con entre 3 y 20 veces menos parámetros; también hubo una variación apreciable entre semillas y no se validó transferencia entre tostadores, recetas o variedades. El estudio modela transferencia de calor y masa, pero no emplea FTIR ni un codificador espectral ([3.1]).

Estos resultados son importantes porque muestran que **añadir física no garantiza automáticamente mayor exactitud, parsimonia o estabilidad**. La ecuación debe corresponder al mecanismo limitante y las variables iniciales deben estar observadas o ser identificables; de lo contrario, la restricción puede introducir sesgo.

Situación por matriz

- **Cacao:** la espectroscopia se concentra en autenticidad, origen, adulteración y calidad sensorial o composicional. No se identificaron series FTIR que conecten fermentación, secado o almacenamiento con una red temporal y ecuaciones de oxidación de polifenoles, pérdida de humedad o formación de compuestos de aroma ([1.1], [2.1]).
- **Café:** es la matriz más avanzada en monitorización de proceso y posee el antecedente PIML-ODE de tostado. Sin embargo, la rama espectroscópica sigue dominada por quimiometría estática. Un estudio FTIR de adulteración con PCA/PLS reportó R^2 interno superior a 0.999, pero advirtió expresamente sobre tamaño muestral limitado, riesgo de sobreajuste y validez restringida a las mezclas estudiadas ([4.1]).
- **Miel:** la evidencia recuperada se orienta a origen botánico y adulteración. Falta modelar conjuntamente HMF, actividad diastásica, humedad, color, fermentación y cristalización como estados dinámicos observables a través de FTIR ([2.2]).
- **Aceite de coco:** la literatura de aceites comestibles confirma que FTIR se usa para clasificación, adulteración, perfil de ácidos grasos y monitorización de oxidación. Sin embargo, no se encontró una integración específica para coco que combine codificación profunda, memoria temporal y cinética de oxidación. Las revisiones señalan dependencia de bibliotecas de referencia, falta de transferencia entre instrumentos y validación insuficiente entre aceites, proveedores, estaciones y laboratorios ([5.1], [5.2], [5.3]).

Vacíos metodológicos principales

Integración multi-matriz

1. **Ausencia de conjuntos longitudinales armonizados.** Faltan espectros repetidos del mismo lote, con temperatura, humedad, oxígeno, luz y envase controlados, acompañados de referencias químicas destructivas.
2. **Heterogeneidad física de las matrices.** Cacao y café son sólidos dispersivos; miel es una fase acuosa concentrada; el aceite de coco es lipídico y puede cambiar de fase. Un preprocesamiento o encoder único podría aprender principalmente diferencias de estado físico en vez de mecanismos compartidos.
3. **Fusión no equivale a generalización multi-matriz.** La combinación de instrumentos o clasificadores dentro de una matriz no demuestra transferencia entre alimentos. Los antecedentes de fusión alimenta-

ria suelen integrar NIR, MIR, Raman, imágenes o cromatografía con PCA/PLS o clasificadores, sin un espacio químico común ni validación *leave-one-matrix-out* ([1.1], [1.2]).

4. **Transferencia instrumental insuficiente.** Los espectros pueden no ser directamente portables entre ATR, transmisión, resoluciones, cristales, fabricantes o laboratorios. La literatura de aceites advierte que los espectros de referencia y modelos pueden no transferirse entre instrumentos ([5.1]).
5. **Validación débil.** Son necesarios particionamientos por lote, origen, campaña e instrumento, no divisiones aleatorias de espectros correlacionados. También faltan pruebas prospectivas y frente a condiciones fuera de distribución.

Restricciones cinéticas

1. No está establecido qué leyes deben imponerse: Arrhenius, oxidación autocatalítica, reacciones consecutivas, difusión limitada, pérdida de humedad, Maillard o cinética microbiana.
2. Las cuatro matrices no comparten necesariamente el mismo mecanismo ni orden de reacción; una Neural ODE jerárquica debería separar parámetros globales, específicos de matriz y específicos de lote.
3. Faltan estudios de **identificabilidad**: distintos mecanismos pueden producir trayectorias espectrales similares.
4. Casi no se cuantifica incertidumbre de parámetros cinéticos ni de vida útil; una predicción puntual puede ser precisa pero químicamente no identificable.
5. No se comparan de manera sistemática restricciones duras, penalizaciones blandas y modelos híbridos residuales. El caso del café muestra que una restricción físicamente motivada puede costar parámetros y estabilidad sin mejorar el resultado ([3.1]).

Interpretabilidad química

Los mapas de atención son útiles para localizar bandas, pero **atención no implica causalidad ni asignación química**. Una banda puede recibir peso por dispersión, humedad, línea base, instrumento o correlación con el lote. La interpretabilidad actual suele limitarse a mapas de atención, SHAP, LIME o proyecciones UMAP; los propios trabajos reconocen comportamiento de caja negra y sensibilidad a variación de muestra ([6.4], [6.5]).

Una interpretación químicamente sólida debería exigir:

- estabilidad de la atribución entre semillas, preprocesamientos, instrumentos y matrices;
- correspondencia con asignaciones de bandas y estándares puros;
- concordancia con análisis ortogonales, como HPLC, GC-MS, índice de peróxidos, HMF, humedad o actividad antioxidante;
- pruebas de ablación por regiones espectrales y perturbaciones físicamente plausibles;
- estados latentes con restricciones de positividad, monotonicidad, balance de masa y estequiometría;
- recuperación de constantes cinéticas conocidas en datos simulados antes de aplicarse a muestras reales.

Arquitectura de referencia para cerrar el vacío

Una evaluación convincente debería comparar: **encoder 1D Transformer frente a GNN de bandas/grupos funcionales**, seguido de **LSTM/GRU frente a Mamba**, y finalmente tres cabezas: regresión libre, Neural ODE y PINN con cinéticas específicas de matriz. El entrenamiento debería ser multitarea para predecir propiedades como humedad, acidez, índice de peróxidos, HMF, polifenoles y marcadores de Maillard, no solamente una etiqueta de edad.

El protocolo mínimo tendría que incluir múltiples lotes, campañas, países, instrumentos y condiciones aceleradas y reales; división externa por lote e instrumento; validación *leave-one-matrix-out*; calibración de incertidumbre; y comparación contra PLS, RF, 1D-CNN y modelos cinéticos clásicos. Sin esos controles, un modelo de gran capacidad podría interpolar edad o procedencia sin aprender degradación química.

En síntesis, existe **viabilidad técnica demostrada por componentes**, pero no evidencia consolidada de la arquitectura integral solicitada. El vacío principal no es simplemente “probar un modelo más profundo”, sino construir datos longitudinales multi-matriz y una representación en la que bandas FTIR, propiedades fisicoquímicas y estados cinéticos sean conjuntamente identificables, transferibles e interpretables.

References

[1.1] Data Fusion for the Food Industry: Leveraging Machine Learning and Deep Learning Algorithms. G Felizzato, E Bagnulo, A Guglielmetti, C Bortolini. 2025. https://iris.unito.it/bitstream/2318/2094551/1/CCM2025_BoA.pdf Context: "The excerpt provides no evidence that cocoa, coffee, honey, or coconut-oil FTIR spectra are integrated with Transformer or GNN spectral encoders, temporal LSTM/GRU/Mamba models, or physics-informed kinetic constraints such as PINNs or Neural ODEs. The cocoa application instead studies off-flavour and industrially accepted liquor using LC-MS and GC-MS, with conventional machine-learning and shallow/deep classification methods: KNN, LDA, logistic regression, PLS-DA, random forest, SVM, and multilayer perceptron. Data fusion is introduced as integration of signals from different instruments, but the supplied text ends before specifying the LLDF/MLDF results, validation design, degradation time course, or mechanistic chemical interpretation. Thus, no time-dependent physicochemical degradation prediction, kinetic modeling, or explicit FTIR modality is established here. Adjacent evidence shows multiblock spectroscopy separately: visible, NIR, and MIR measurements of fat samples are analyzed with PCA, consensus PCA, and SO-PLS, with NIR reported as the best predictor of fat content. This supports multimatrix/multisensor integration and exploratory interpretability, but not Transformer/GNN, recurrent temporal, or physics-informed architectures. The Raman example reports Monte Carlo cross-validation and an external test set, while the cocoa excerpt gives no validation metrics in the supplied portion."

[1.2] Data Fusion for the Food Industry: Leveraging Machine Learning and Deep Learning Algorithms. G Felizzato, E Bagnulo, A Guglielmetti, C Bortolini. 2025. https://iris.unito.it/bitstream/2318/2094551/1/CCM2025_BoA.pdf Context: "The excerpt provides adjacent evidence for spectroscopy-based food-process monitoring, but not for the requested integrated FTIR–Transformer/GNN–temporal neural architecture with physics-informed kinetic constraints. For coffee, NIR spectra were collected from green beans, roasted coffee, and capsules across production months and analyzed using preprocessing, SNV correction, mean centering, PCA, and selected spectral intervals. The study associated spectral variation with lipid, protein, and moisture differences linked to industrial processing; environmental effects were detectable in roasted coffee and final products but not green beans. For table olives, MIR and Raman spectra monitored 145 days of fermentation under varying NaCl/KCl substitution. PCA, ASCA, covariance analysis, and PLS were used to identify time and salt effects and predict pH and titratable acidity. A separate plant-food shelf-life study combined RGB and NIR hyperspectral imaging with MCR-ALS, pH, and weight loss over storage at 5, 10, and 20°C, extracting spectral, spatial, and temporal degradation information. These methods demonstrate multimodal/multimatrix measurement, conventional validation through prediction and variability analysis, and some chemical interpretability, but the excerpt reports no FTIR cocoa, honey, or coconut-oil study, no Transformer/GNN, LSTM/GRU/Mamba, PINN, or Neural ODE, and no explicit mechanistic kinetic constraint."

[2.1] Developing a novel onsite detection technology by using chemometrical analysis of hand-held near-infrared sensor technique for assessing coffee quality. VG Boadu. 2024. <https://ir.ucc.edu.gh/server/api/core/bitstreams/dfa14ec2-379d-4097-b588-070784b92769/content> Context: "This excerpt is a bibliography rather than a report of experiments, so it does not provide evidence that FTIR spectra from cocoa, coffee, honey, or coconut oil have been integrated with Transformer/GNN spectral encoders, temporal LSTM/GRU/Mamba models, or physics-informed kinetic constraints such as PINNs or Neural ODEs. The cited studies instead establish conventional adjacent methods and tasks: near-infrared monitoring of coffee roasting for sucrose and colour; NIR measurement of chlorogenic acid; classification of Brazilian coffee; FT-NIR discrimination of Ghana cocoa and quantification of cocoa adulteration; and coffee adulteration/authenticity analysis using chemometrics. Other references concern acrylamide, aroma formation, microbial/fungal processes, and roasting impacts, but the excerpt does not describe time-series model architectures, degradation trajectories, multimatrix fusion, kinetic equations, uncertainty validation, or chemically interpretable latent representations. Neural networks appear separately in a cited study for online prediction of fermentation variables, while spectroscopy papers primarily use multivariate calibration, classification, SIMCA, PLS-DA, or chemometrics. Thus,

the references support the existence of spectroscopy-based food-quality prediction and process-monitoring components separately, but not their claimed combination or validation for time-dependent physicochemical degradation. No honey or coconut-oil Transformer/GNN/PINN evidence is visible here."

[2.2] Developing a novel onsite detection technology by using chemometrical analysis of hand-held near-infrared sensor technique for assessing coffee quality. VG Boadu. 2024. <https://ir.ucc.edu.gh/server/api/core/bitstreams/dfa14ec2-379d-4097-b588-070784b92769/content> Context: "This excerpt is a bibliography rather than experimental evidence for an integrated FTIR/Transformer-GNN/temporal/PINN framework. It documents adjacent spectroscopy applications, chiefly for coffee and honey. Reported tasks include coffee integrity and husk-adulteration detection, Arabica/Robusta or processing-form authentication, caffeine prediction, roasted-coffee blend quantification, roasting-process monitoring, and antioxidant-property estimation. Methods explicitly represented are classical chemometrics (PLS, MLR, PCA, multivariate statistical analysis, discriminant analysis, and random forests), FTIR/NIR spectroscopy, electronic nose/tongue combinations, and one CNN-plus-FT-NIR adulteration study. The references therefore support spectral classification, compositional prediction, and some real-time roasting or quality monitoring, but do not provide evidence here for Transformer or GNN spectral encoders, LSTM/GRU/Mamba temporal models, Neural ODEs, PINNs, kinetic degradation constraints, or unified multimatrix modeling across cocoa, coffee, honey, and coconut oil. Validation details, longitudinal degradation experiments, external testing, and chemically interpretable temporal mechanisms are not available in this excerpt. Chemical interpretability is only indirectly suggested through analytes such as caffeine, trigonelline, nicotinic acid, antioxidant properties, and trace-element ratios. Adjacent evidence establishes that spectroscopy, chemometrics, CNNs, and real-time food-process monitoring exist separately, but not their proposed integration."

[3.1] Neural-network placement in physics-informed machine learning for mechanistic process-model repair: a case study in industrial coffee roasting. Morgen Pronk, Brian W. Anthony. Scientific Reports (2026). <https://doi.org/10.1038/s41598-026-67034-7> Context: "This excerpt provides peer-reviewed, 2026 Article-in-Press evidence on physics-informed machine learning for an industrial food process: coffee roasting. Pronk and Anthony compare four neural-network placement strategies around a mechanistic ordinary differential equation (ODE), plus a matched-input feedforward multilayer perceptron (MLP) baseline, using 221 roasts. The PIML approaches improve autonomous rollout performance from a poor baseline (inline-eq-IEq1 = -0.44) to 0.70–0.94, while the non-mechanistic matched-input neural model reaches 0.97 with 3–20 times fewer parameters. PIML models share difficult roasts, whereas the baseline has a structurally different error profile. A physics-motivated bound on residual corrections is not consistently beneficial: the unbounded model uses 3.7 times fewer parameters and improves one reported metric by 0.02. Optimization stability also differs substantially across placements, with seed variation spanning 0.009–0.117. The mechanistic scaffold is based on heat- and mass-transfer literature for bean-temperature evolution, but the authors frame missing run-specific initialization—not necessarily incorrect physics—as a major source of rollout failure. Scope is limited to one industrial roaster and three recurring roast labels; transfer across machines, recipes, or bean varieties was not validated. The excerpt contains no evidence about Mamba, FTIR/chemometrics, edible oil, honey, cocoa, spectroscopy, multimodal fusion, or external validation beyond this within-configuration coffee study."

[4.1] Detection of Adulteration in Coffee Products Using FTIR Spectroscopy and Multivariate Analysis. Sonny Widiarto, La Zakaria, Safitri Hauratul Jamal, Dian Septiani Pratama, Irwan Saputra. Jurnal Kimia Sains dan Aplikasi (2026). <https://doi.org/10.14710/jksa.29.4.296-305> Context: "This excerpt provides evidence for conventional food-spectroscopy methodology in coffee authentication, specifically FTIR combined with PCA and PLS. FTIR is explicitly involved and is described as efficient, rapid, and cost-effective. The study used reference standards, market samples, calibration and validation sets, and chemometric processing. PCA indicated that samples A–C clustered with robusta, while D–E exhibited spectral characteristics associated with corn powder; corn was inferred to be the dominant adulterant over rice. A PLS model was then developed for robusta–corn mixtures, with reported internal R² greater than 0.999 and estimated robusta contents of approximately 71.7%–98.2%. However, the text explicitly identifies the work as preliminary and warns that the limited sample size and calibration design may cause overfitting, making predictions specific to the studied mixtures and conditions. The

excerpt contains no information about ASGMamba or Mamba/SSM computational properties, Perré 2025 drying PINN or physics-informed kinetic modeling, edible-oil authenticity reviews, or a general food-spectroscopy review. It concerns coffee only; no evidence here establishes involvement of four additional target foods. Thus, FTIR and coffee are involved, while the other requested methods, review texts, and target foods are not represented in this excerpt."

[5.1] Review of Current and Emerging Analytical Methods for the Testing of Edible Oil for Authenticity. Helen H. Grundy, Hez J. Hird, Victoria Bailey-Horne, M. Sykes, A. Charlton. FSA Research and Evidence (2025). <https://doi.org/10.46756/001c.145026> Context: "This excerpt provides evidence about conventional spectroscopy for edible-oil authenticity, especially infrared methods and FTIR. It describes IR as rapid and non-destructive, but dependent on characteristic fingerprints, extensive reference databases, and multivariate classification models. The review identifies important validation and standardization problems: spectral interpretation is effectively a "black box," reference spectra may not transfer between instruments, and environmental and instrument-specific factors can affect measurements. FTIR is explicitly involved: it is reported as robust for detecting total sterols in extra-virgin olive oil, although it gave less detail about individual sterol changes. MIR/ATR is also reported for detecting palm-oil adulteration in sunflower oil, with a minimum reported detection level of 5%; NIR applications include moisture, fatty-acid, sterol, and geographical-origin measurements. The excerpt also states that NMR proof-of-concept studies for avocado, sesame, grape-seed, and fish oils require further validation before wider use. It contains no evidence about Mamba architectures, state-space models (SSMs), ASGMamba, Perré's 2025 drying PINN, physics-informed kinetic modeling, or coffee FTIR studies. Among the named material, only the edible-oil review is represented here. FTIR and edible oils are involved; the four target foods are not identifiable or evidenced in this excerpt."

[5.2] Review of Current and Emerging Analytical Methods for the Testing of Edible Oil for Authenticity. Helen H. Grundy, Hez J. Hird, Victoria Bailey-Horne, M. Sykes, A. Charlton. FSA Research and Evidence (2025). <https://doi.org/10.46756/001c.145026> Context: "This excerpt provides evidence about FTIR and related infrared spectroscopy for edible-oil authenticity, but it contains no information about ASGMamba, Mamba or state-space-model (SSM) computational properties, and no physics-informed neural-network (PINN) kinetic modeling. FTIR is explicitly involved: the review describes it as effective for edible-oil analysis, adulterant detection, fatty-acid measurement, classification, geographical-origin assessment, and oxidation monitoring. It also reports FTIR discrimination among 17 oil types and applications to olive-oil adulteration. Validation and standardization limitations are apparent. A preliminary visible/near-infrared study classified EVOO and VOO with 82.35% and 66.67% correct classification, respectively, and the authors requested broader, more numerous samples and IOC-organized future studies. The review characterizes spectroscopy as suitable for rapid, non-destructive preliminary screening, with unsatisfactory samples subsequently tested by official methods. Conventional analytical methods are referenced generally, especially chromatography and official analytical methods; spectroscopy is described as potentially faster, cheaper, and less reliant on toxic reagents. The excerpt concerns edible oils and mentions crisps/potato chips in a frying-oil application, but it does not identify the question's four target foods as a defined set. Thus, only edible oil—and incidentally potato chips—is explicitly represented here."

[5.3] Review of Current and Emerging Analytical Methods for the Testing of Edible Oil for Authenticity. Helen H. Grundy, Hez J. Hird, Victoria Bailey-Horne, M. Sykes, A. Charlton. FSA Research and Evidence (2025). <https://doi.org/10.46756/001c.145026> Context: "This excerpt provides evidence about edible-oil authenticity testing, but not about ASGMamba, Mamba/SSM computational properties, or Perré 2025 physics-informed neural-network kinetic modeling. It describes rapid screening approaches, mainly spectroscopy, alongside conventional or confirmatory methods based on fatty-acid, sterol, and triacylglycerol profiling. FTIR is explicitly involved: Fourier Transform Infrared spectroscopy is identified as a promising rapid, low-cost, point-of-use method, although further validation is required. Raman spectroscopy is also mentioned. The review highlights major validation and standardization problems: many methods are immature, insufficiently tested across diverse oils, suppliers, sources, and seasonal conditions, and have not undergone collaborative trials. It also reports limited proficiency-testing uptake, accuracy concerns, proprietary business protocols, and inadequate regulation and standardization. Triacylglycerol analysis is presented as a potentially accessible confirmatory method for official

laboratories, while certified reference materials are recommended to support quality control. The excerpt concerns edible oils only; it does not identify the four target foods in the question, and it contains no evidence connecting FTIR or these validation issues to those foods. No image evidence was provided or used."

[6.1] Enhancing spectral analysis of ghee adulteration via a deep learning-based multimodal attention mechanism. Aditya Upadhyay, Neha Chaudhary. Scientific Reports (2026). <https://doi.org/10.1038/s41598-026-48661-6> Context: "The excerpt provides adjacent evidence for FTIR-based deep learning in food analysis, but it does not establish the proposed cocoa, coffee, honey, or coconut-oil degradation framework. The study concerns ghee adulteration detection and food authentication, not prediction of time-dependent physicochemical degradation. Its architecture combines conventional classifiers (including SVM, XGBoost, and Random Forest) with an attention-augmented 1D-CNN and a Transformer network. A Spectral Attention Map is intended to identify chemically relevant FTIR bands and improve interpretability, while Dynamic Multimodal Fusion adaptively weights model outputs. The reported evaluation uses a custom, statistically enriched FTIR dataset with different ghee attributes and reports 99.6% classification accuracy after 94% feature-dimensionality reduction. Thus, the excerpt supports the existence of Transformer-based spectral encoding, attention-based chemical-band selection, multimodal fusion, and FTIR food-authentication tasks. However, it gives no evidence of temporal LSTM, GRU, or Mamba models; physics-informed neural networks, Neural ODEs, or kinetic constraints; longitudinal degradation prediction; or integration across cocoa, coffee, honey, and coconut-oil matrices. It also does not demonstrate multimatrix validation or chemical interpretability beyond attention maps. Because the manuscript is an unedited article-in-press version, its claims should be treated cautiously pending final editing and fuller methodological and validation details."

[6.2] Enhancing spectral analysis of ghee adulteration via a deep learning-based multimodal attention mechanism. Aditya Upadhyay, Neha Chaudhary. Scientific Reports (2026). <https://doi.org/10.1038/s41598-026-48661-6> Context: "This excerpt provides adjacent evidence for FTIR-based food authentication, but it does not evaluate time-dependent physicochemical degradation in cocoa, coffee, honey, or coconut oil. The actual task is classifying ghee adulteration, including undeclared animal fats, rather than predicting degradation trajectories or kinetics. The proposed multimodal system combines traditional classifiers (Random Forest, SVM, and XGBoost) with an attention-augmented CNN and Transformer through confidence-weighted Dynamic Multimodal Fusion (DMFM). It reports accuracy above 95%, but the excerpt does not establish external, longitudinal, or multimatrix validation. Spectral Attention Maps (SAM) rank chemically informative FTIR bands, specifically ester linkages near 1740 cm⁻¹ and C–H stretching near 2850 cm⁻¹, reducing feature space by approximately 40%; attention heatmaps and UMAP are used for interpretability. Raspberry Pi deployment supports real-time inference. The literature discussion notes that CNNs and Transformers can model local and long-range spectral dependencies, but no Graph Neural Network, LSTM, GRU, Mamba, PINN, Neural ODE, or explicit kinetic constraint appears. Thus, the excerpt supports separate existence of Transformer-based spectral encoding, multimodal fusion, and chemically informed attention, while providing no evidence that these are integrated with temporal sequence models or physics-informed degradation modeling. It also cautions that reported high accuracies may decline on novel adulterants and real-world field data."

[6.3] Enhancing spectral analysis of ghee adulteration via a deep learning-based multimodal attention mechanism. Aditya Upadhyay, Neha Chaudhary. Scientific Reports (2026). <https://doi.org/10.1038/s41598-026-48661-6> Context: "This excerpt concerns a peer-reviewed 2026 Scientific Reports proposal and reported framework for FTIR-based ghee adulteration detection; it does not provide evidence about PINN/Neural ODE food-kinetics models, Mamba architectures, or applications to coffee, honey, cocoa, or other edible oils. Its conventional spectroscopy context is food-authentication classification using FTIR, with earlier chemometric/machine-learning methods described as effective for fats and oils but increasingly limited as adulteration becomes more sophisticated. The paper proposes a multimodal system combining Random Forest, SVM, XGBoost, an attention-augmented CNN, and a Transformer. Its Spectral Attention Map (SAM) automatically ranks chemically informative bands, including ester linkages near 1740 cm⁻¹ and C–H stretching near 2850 cm⁻¹, reportedly reducing feature space by about 40% without major performance loss. Its Dynamic Multimodal Fusion Mechanism (DMFM) weights model predictions using online confidence scores and reports accuracy above 95%, exceeding individual

models. Interpretability is addressed through attention heatmaps and UMAP, while Raspberry Pi deployment targets real-time edge inference. Important validation limitations are explicitly acknowledged: high reported accuracies may fall on novel adulterants, real-world field data, or statistically enhanced internal datasets. The excerpt does not establish external validation, standardized datasets/protocols, or comparison across target matrices, and the SAM/DMFM results should be treated as claims from this 2026 work rather than established consensus."

[6.4] Enhancing spectral analysis of ghee adulteration via a deep learning-based multimodal attention mechanism. Aditya Upadhyay, Neha Chaudhary. Scientific Reports (2026). <https://doi.org/10.1038/s41598-026-48661-6> Context: "This excerpt does not provide evidence that FTIR spectra of cocoa, coffee, honey, or coconut oil were integrated with Transformer/GNN encoders, temporal LSTM/GRU/Mamba models, or physics-informed kinetic constraints such as PINNs or Neural ODEs to predict time-dependent physicochemical degradation. Its demonstrated application is principally ghee authentication/adulteration detection, using FTIR alongside classical chemometrics, SVM, RF, XGBoost, and related deep-learning approaches. The article identifies FTIR as rapid and non-destructive, while contrasting it with accurate but costly and slow HPLC and GC-MS. It reports adjacent use of CNNs and Transformers for local and long-range spectral features, a hybrid Transformer-CNN for soil analysis, and multimodal Raman-plus-FTIR fusion for improved analytical capacity in cancer-related tasks. Attention maps, SHAP, and LIME are cited for chemical-region interpretability, although interpretability remains a limitation of conventional machine-learning and deep-learning models. The excerpt also notes deployment challenges: complex models require more data and computation, hybrid models are difficult to deploy on portable devices, and FTIR studies often suffer from sample variation and black-box behavior. No time-series degradation experiment, kinetic validation, multimatrix cocoa/coffee/honey/coconut dataset, GNN, recurrent/state-space temporal model, PINN, Neural ODE, or prospective process validation is described."

[6.5] Enhancing spectral analysis of ghee adulteration via a deep learning-based multimodal attention mechanism. Aditya Upadhyay, Neha Chaudhary. Scientific Reports (2026). <https://doi.org/10.1038/s41598-026-48661-6> Context: "This excerpt primarily provides peer-reviewed background and research-gap evidence for FTIR-based ghee authentication; it does not report PINN/Neural ODE food-kinetics models, Mamba architectures, or applications to coffee, honey, cocoa, or other edible-oil matrices. Conventional laboratory methods such as HPLC and GC-MS are described as accurate, but poorly suited to field deployment because of equipment cost, skilled-labor requirements, sample preparation, and slow turnaround. FTIR is presented as a rapid, non-destructive alternative, although existing FTIR workflows largely rely on chemometrics and classical machine learning. Their limitations include difficulty modeling high-dimensional nonlinear spectral relationships, sensitivity to sample variation, and weak interpretability, which can reduce regulatory acceptance and confidence. The cited literature indicates that CNNs and Transformers can capture local features and long-range spectral dependencies, while hybrid Transformer-CNN designs may improve performance over separate models. Raman-FTIR fusion is identified as a complementary multimodal strategy that can increase analytical capacity, though data-fusion complexity remains a disadvantage. SHAP, LIME, and attention heatmaps are proposed or used for spectral interpretability. Edge deployment on inexpensive hardware, including Raspberry Pi, is reported as enabling real-time testing with decision times under ten seconds. No evidence here addresses external validation or standardization in detail."

[7.1] ASGMamba: Adaptive Spectral Gating Mamba for Multivariate Time Series Forecasting. Qian-Yang Li, Xingjun Zhang, Shao-Xun Wang, Jia Wei, Yueqi Xing. ArXiv (2026). <https://doi.org/10.48550/arxiv.2602.01668> Context: "The excerpt provides evidence specifically about ASGMamba's computational and spectral-filtering properties. It contrasts Transformer models' quadratic sequence-length complexity, $O(L^2)$, with Mamba's strict linear scaling, $O(L)$, enabled by a hardware-aware parallel scan. It states that standard SSMs can struggle with high-frequency noise, which may consume latent-state capacity and reduce the model's ability to represent long-term trends. ASGMamba addresses this with Adaptive Spectral Gating (ASG), an input-dependent, local frequency-selective filter that uses local patch spectral energy to suppress noise before or during state evolution. FFT is applied only to fixed-size patches, preserving $O(L)$ complexity rather than the $O(L \log L)$ cost associated with global Fourier transforms. The article also claims reduced memory usage and latency for long-horizon forecasting, while retaining a hierarchical multi-scale design and variable-specific Node Embeddings. This excerpt

contains no evidence about Perré 2025 drying PINNs, physics-informed kinetic modeling, edible-oil authenticity, coffee FTIR analysis, general food-spectroscopy reviews, validation or standardization problems, or conventional food-analysis methods. FTIR is not mentioned, and none of the four target foods is identified in the supplied ASGMamba pages."

[8.1] Toward mechanistic models "augmented" by machine learning. Example of a drying simulation data set exploited by a Physics Informed Neural Network. Patrick Perré. Drying Technology (2025). <https://doi.org/10.1080/07373937.2024.2411582> Context: "This excerpt provides evidence about physics-informed machine-learning modeling in drying, but it does not discuss ASGMamba, state-space/Mamba computational properties, FTIR, edible-oil authenticity, coffee spectroscopy, or the four target foods. Perré describes mechanistic drying models as being based on fundamental balance equations and relevant material parameters, with the ability to predict configurations not previously observed. Their principal limitation is operational complexity, particularly when representing the complete set of product characteristics. Machine-learning methods can handle nonlinear and dynamic situations, but are constrained by the domain represented in their training data. The proposed strategy is to combine both approaches into an "augmented" mechanistic model through fully coupled, fully decoupled, or cascade coupling. In the demonstrated fully decoupled approach, the mechanistic model generates a simulation dataset and a neural network subsequently exploits it; the study compares a conventional neural network with a physics-informed neural network. The application concerns coupled heat and mass transfer in drying, including moisture-content and temperature fields, rather than kinetic modeling of food authenticity. The excerpt also does not provide validation or standardization evidence, nor does it describe conventional analytical methods such as FTIR."

[9.1] Convolutional neural networks in Vis–NIR chemometrics: From contradiction to conditional design. Dário Passos. TrAC Trends in Analytical Chemistry (2605). <https://doi.org/10.1016/j.trac.2026.119015> Context: "This excerpt is a bibliography rather than an evidence synthesis, so it does not establish that FTIR spectra of cocoa, coffee, honey, or coconut oil have been integrated with Transformer or GNN spectral encoders, temporal LSTM/GRU/Mamba models, and physics-informed kinetic constraints such as PINNs or Neural ODEs for time-dependent physicochemical degradation. The closest matrix-specific evidence is a CNN combined with FT–NIR for coffee adulteration, which addresses classification/prediction rather than degradation kinetics. Adjacent spectroscopy studies use one-dimensional CNNs for spectroscopic regression, CNNs for vibrational spectroscopy, CNNs for NIR calibration, end-to-end quantitative spectral analysis, dilated CNNs, residual networks, multi-output CNNs, and CNN-based spectral preprocessing. A CNN–LSTM example exists for infrared identification of pre-cooked pasta in different physical states, but this is not evidence of kinetic degradation modeling or the use of GRU/Mamba, Transformers, GNNs, PINNs, or Neural ODEs. The references also indicate adjacent concerns involving transferability between spectrometers, preprocessing, and simultaneous prediction of multiple traits. However, the excerpt provides no reported validation design, multimatrix integration, temporal sampling, mechanistic constraints, uncertainty analysis, or chemical interpretability results. It therefore supports the existence of separate CNN, LSTM, FT–NIR, transfer-learning, and food-classification components, but not their proposed unified architecture."